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Architecting Agentic AI Solutions

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Agenda

Logical Architectural of Agentic AI Solutions

- Logical Architecture Diagram
  - LLM Router
  - Tools Registry
  - Tools Gateway
  - Integration with Legacy System

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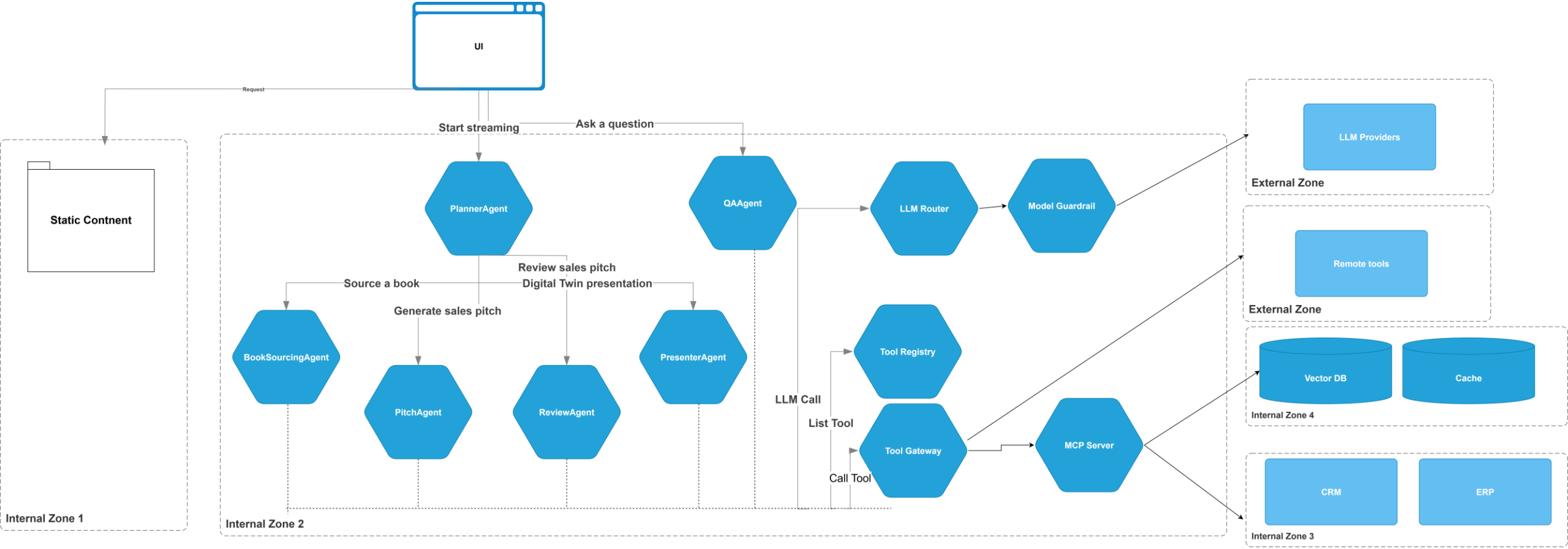
Logical Architecture of Agentic AI Solution

Logical Architecture Diagram



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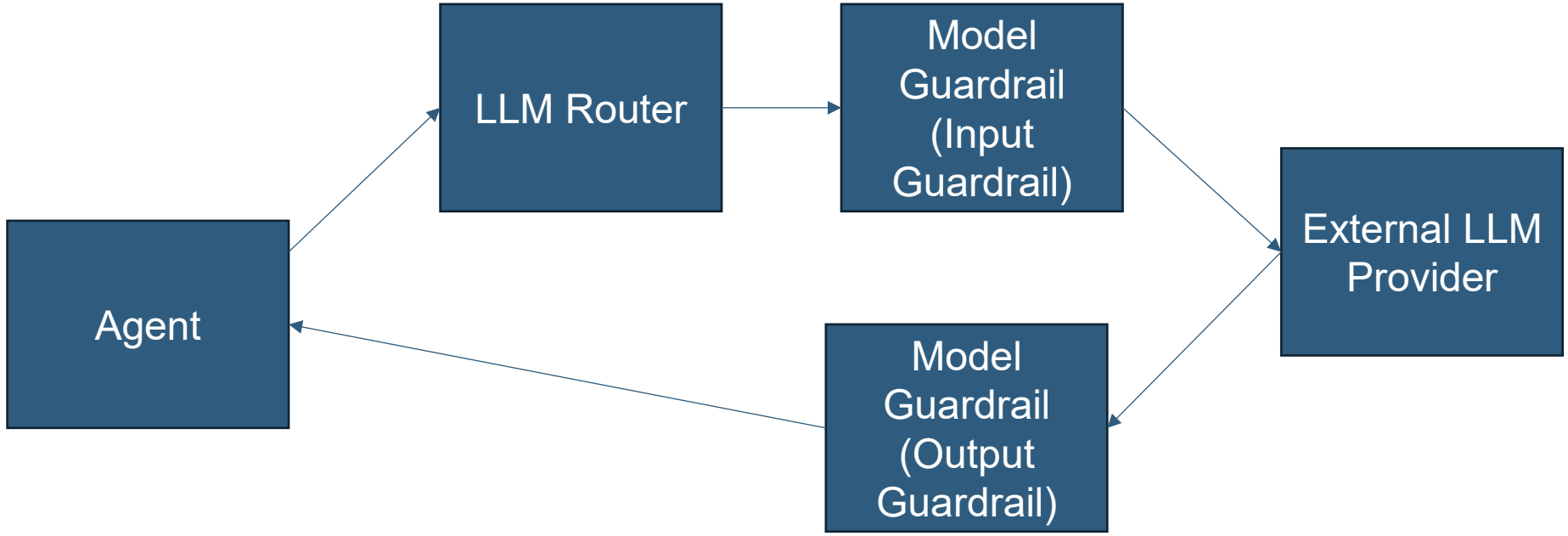
Agent LLM Call Flow



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LLM Router

LLM Router plays the role of a **traffic controller** for all requests that need to go to large language models (LLMs). Its main purposes are:

1. Routing to the Right LLM
Different tasks may require different LLMs. The router decides which LLM provider or model is best suited for a given request, based on:

- Task type (reasoning vs. creative vs. factual).
- Cost vs. quality trade-off.
- Latency/SLA requirements.

2. Abstraction Layer

- Provides a **unified interface for agents** so they don't need to know about individual LLM APIs. **Can onboard new LLM without impact agents.**
- Agents just say "LLM Call," and the router handles provider details, **API keys**, **rate limiting**, and failover.

3. Scalability & Optimization
Can implement strategies like:

- Fallback (if one LLM fails, try another).
- Hybrid execution (use a smaller, cheaper model first, escalate to a more powerful one if needed).
- **Load balancing** across providers.

4. Performance and Cost

- Semantic caching
- Centralizing model access allows control of cost, quality, and safety.

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Model Guardrail

Input Guardrail

Risk Mitigation

- Prevents prompt injection / adversarial prompts.
- Stops malicious inputs before they hit the LLM.

Policy Enforcement/ Apply enterprise rules

- Example: block disallowed topics (e.g., medical diagnosis, financial advice).
- Ensure prompts align with enterprise policies before calling LLM.

Mask sensitive info (e.g., anonymize user IDs).

Output Guardrail
The raw LLM response is intercepted and validated:

Safety Filtering

- Remove/modify toxic or unsafe outputs (hate speech, PII, harmful instructions).

Policy Enforcement/ Apply enterprise rules

- Ensure generated responses follow compliance rules (e.g., no competitor promotion).
- Restrict output languages if needed.

Consistency & Reliability

- Normalize response formats (JSON, markdown, schemas).
- Repair malformed outputs so downstream agents can parse safely.

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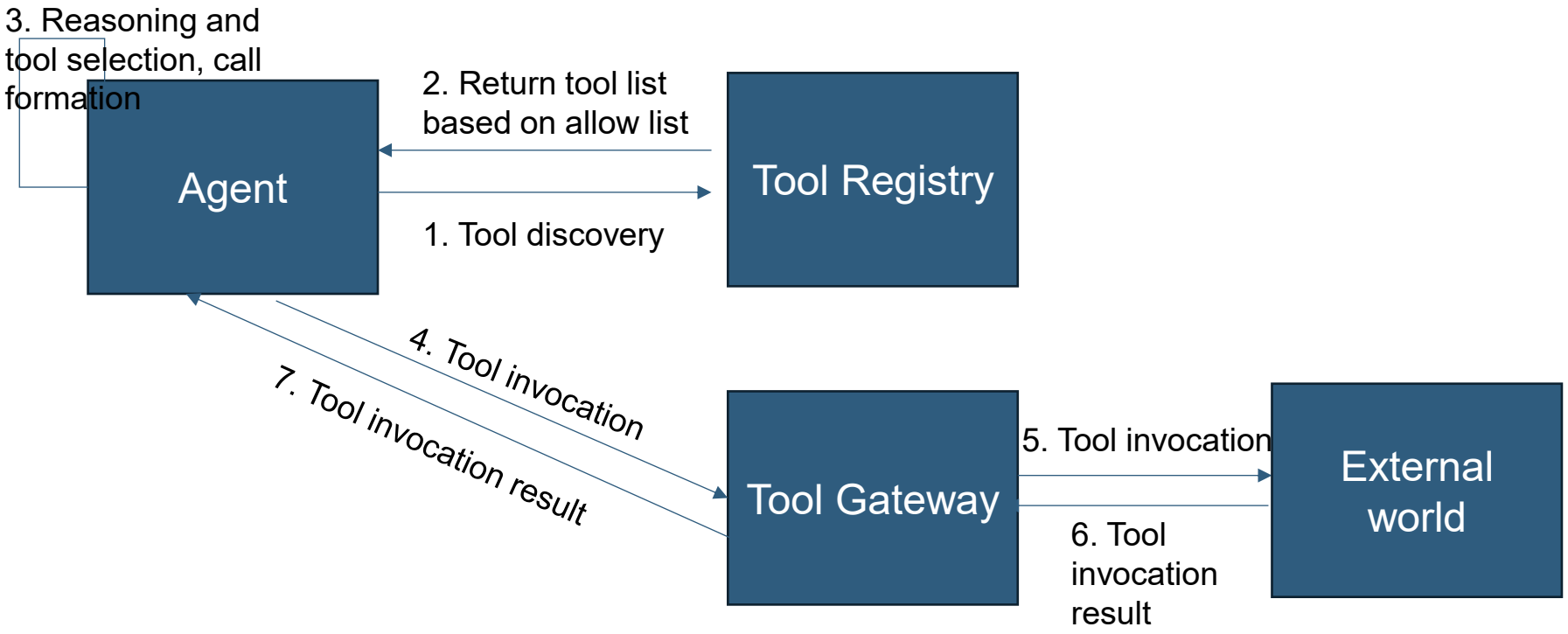
How Agents Dynamically Find & Use Tools



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Tool Registry

Agents need to **discover tools dynamically** instead of being hard-coded. This enables extensibility and governance.

Functionality: Lists available tools, their schemas, versions, and metadata. Helps agents understand *what exists* and *how to call it*.

- The agent calls the tool registry

Registry returns a list of available tools, each with:

  - **Name/ID** (`search`, `erp.transfer`)
  - **JSON Schema** (input/output types)
  - **Metadata** (version, owner, cost, tags, SLOs, deprecation)
- Can be enhanced with role allowed list
- Cache the discovery list with a short TTL (e.g., 60–300s) so agents don't hammer the registry.

Agent → **Registry** = "What tools can I use and how?"

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Tool Gateway

Direct tool calls are risky (security, governance). Gateway ensures all access is controlled and observable.

Functionality: Validates schemas, enforces quotas, applies guardrails, routes to correct MCP server/adapter, logs/traces calls.

- When the agent wants to use a tool, it calls the **internal tool gateway**:
  - `POST /mcp/call` (for MCP)
  - or `POST /tools/<name>` (for plain HTTP adapters)
- The **gateway** enforces runtime policies:
  - AuthN/AuthZ (tenant, role)
  - Schema validation (arguments match JSON Schema)
  - Guardrails (PII scrub, allow/deny lists)
  - Rate/quota/concurrency enforcement
  - Routing (which MCP server or adapter actually handles it)
  - Tracing/metrics injection

This ensures every tool call is **safe, observable, and policy-compliant**, regardless of which agent calls it.

Agent → Gateway = “Please call this tool for me safely.”

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Tool Registry

1.MCP servers as live registries
Each MCP server exposes its own tool list; your **internal gateway** aggregates and filters by tenant/role.

2.Central service catalog
A single microservice stores **all tool metadata** (schemas, owners, versions). Agents call the catalog; **enforces policy** before returning entries.

3.Gateway-native catalog
Some gateways (e.g., Kong + Dev Portal) can serve as the **catalog** with docs and auth baked in. Agents/engineers browse, agents fetch filtered lists at runtime.

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Demo MCP Tool Registry

https://github.com/daveebbelaar/ai-cookbook/tree/main/mcp/crash-course/4-openai-integration

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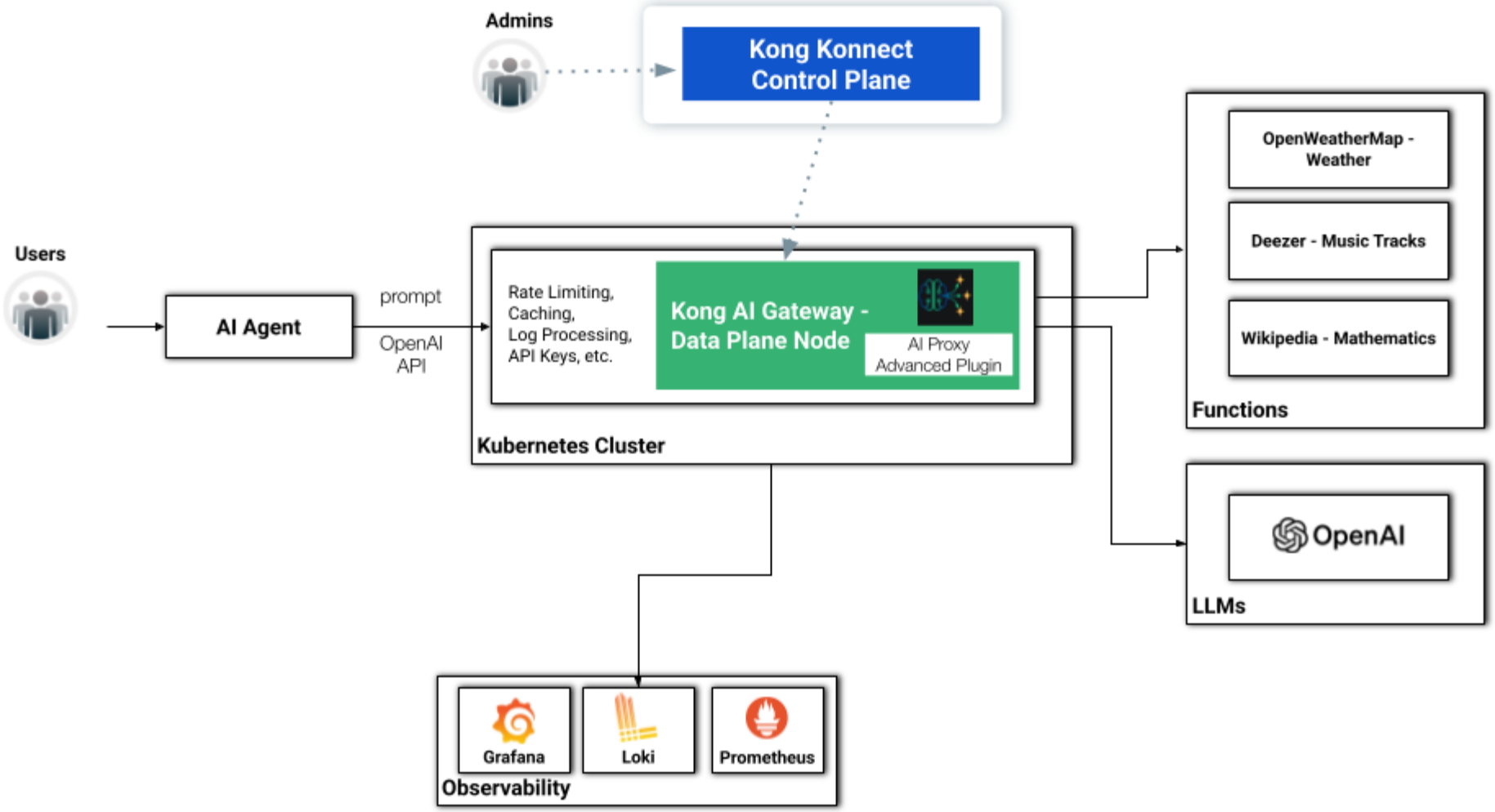
Example of everything in one place



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Guardrails

Policy Guardrails: block PII leakage, enforce allow/deny lists, validate JSON.

Model Guardrails: detect toxicity, hallucinations, unsafe topics, and repair/refuse responses.

Safety gates at the right moments

- **Pre-prompt**: DLP/PII redaction, intent classification, risk scoring
- **Pre-tool**: policy check (allow-list domains, rate limits, per-tool IAM), budget/cost caps
- **Post-tool**: schema validation, sanity checks (evidence count, thresholding)
- **Pre-action (risky ops)**: **HITL approval** (explain plan, diffs, cost-to-date)
- **Post-response**: redact, watermark, log + trace + evidence snapshot
- **Per-tool identities** (cloud IAM/service accounts)
- **Network egress allow-lists** enforced at infra
- **Undo/compensation** pathways for critical actions

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Guardrails Cheat-sheet

Capability	Open-source	AWS	Azure	Google Cloud
PII & DLP	Presidio; regex/NER	Comprehend PII; Macie (at-rest); Bedrock Guardrails (PII)	AI Content Safety (PII); Purview DLP	Cloud DLP
Content safety	Llama Guard; Detoxify	Bedrock Guardrails	Azure AI Content Safety	Vertex/Gemini Safety Settings
Schema & I/O validation	Guardrails AI; NeMo Guardrails; JSON Schema; Pydantic	API Gateway/Step Functions validators; Bedrock Guardrails (resp)	Prompt Flow validators; Azure AI Safety	Grounded Generation safety; Vertex Extensions validators
Prompt/tool injection	Rebuff; Llama Guard; allow-lists	Bedrock Guardrails (jailbreak)	Content Safety jailbreak; APIM policies	Vertex Safety filters; input content filters
Policy-as-code	OPA/Rego; Cedar (OSS)	Verified Permissions (Cedar); IAM; Bedrock Guardrails policies	Azure RBAC; Entra ID; Azure Policy; APIM policies	Cloud IAM Conditions; Policy Controller (Gatekeeper); Org Policy
Sandboxing & egress	gVisor; Firecracker; seccomp/AppArmor; sidecar proxies	VPC, PrivateLink, SG/NACL; ECS/EKS isolation	VNet, Private Link, NSGs; Container Apps/Kubernetes	VPC-SC, Private Service Connect; firewall rules; Cloud Run/VPC
Rate limits & budgets	Envoy/NGINX RL; Redis tokens; OTel cost tags	API Gateway throttles; Service Quotas; Billing Budgets	API Management quotas; Monitor budgets	API Gateway quotas; Service Control; Billing Budgets
Audit & trace	OpenTelemetry; Jaeger; ELK/EFK	CloudTrail; CloudWatch Logs; X-Ray	Azure Monitor; App Insights; Activity Log	Cloud Logging; Cloud Trace; Audit Logs

https://medium.com/agenticai-the-autonomous-intelligence/agentic-ai-architecture-a-practical-production-ready-guide-2b2aa6d16118

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MCP Server

Standardized Connector

- Provides a **uniform protocol (Model Context Protocol)** for agents to interact with external systems.
- Agents don't need to know the system's native API (e.g., SQL, SAP, Salesforce). They just speak MCP.

Abstraction of External Systems

- Each MCP Server acts like a wrapper/adapter for one or more external systems.
- Example: a "CRM MCP Server" exposes customer records in a standardized way, regardless of whether the backend is Salesforce, Dynamics, or a custom database.

Context Packaging

- MCP servers prepare and format domain-specific context (records, actions, workflows) so it's consumable by LLMs/agents.

Extensibility

- New external tools can be added.
- No need to rewire agents or the tool gateway — they automatically benefit from the new capability.

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API Gateway

Put an API gateway in front (WAF, TLS, CORS) and terminate auth with **JWT/OAuth/OIDC**.

Enforce **rate limits** and **quotas** by tenant, user, and agent. Add burst protection and graceful 429s.

Apply **request budgets** (tokens, latency, cost) at the edge; reject over-sized payloads early.

Stamp every request with a **correlation ID**; propagate to all downstream spans and logs.

Define hard **timeouts** and **circuit breakers** to keep failing components from cascading.

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Agent Discovery

Agent Registry

Agent as tool
-> tool registry

Agent as service
-> service
discovery

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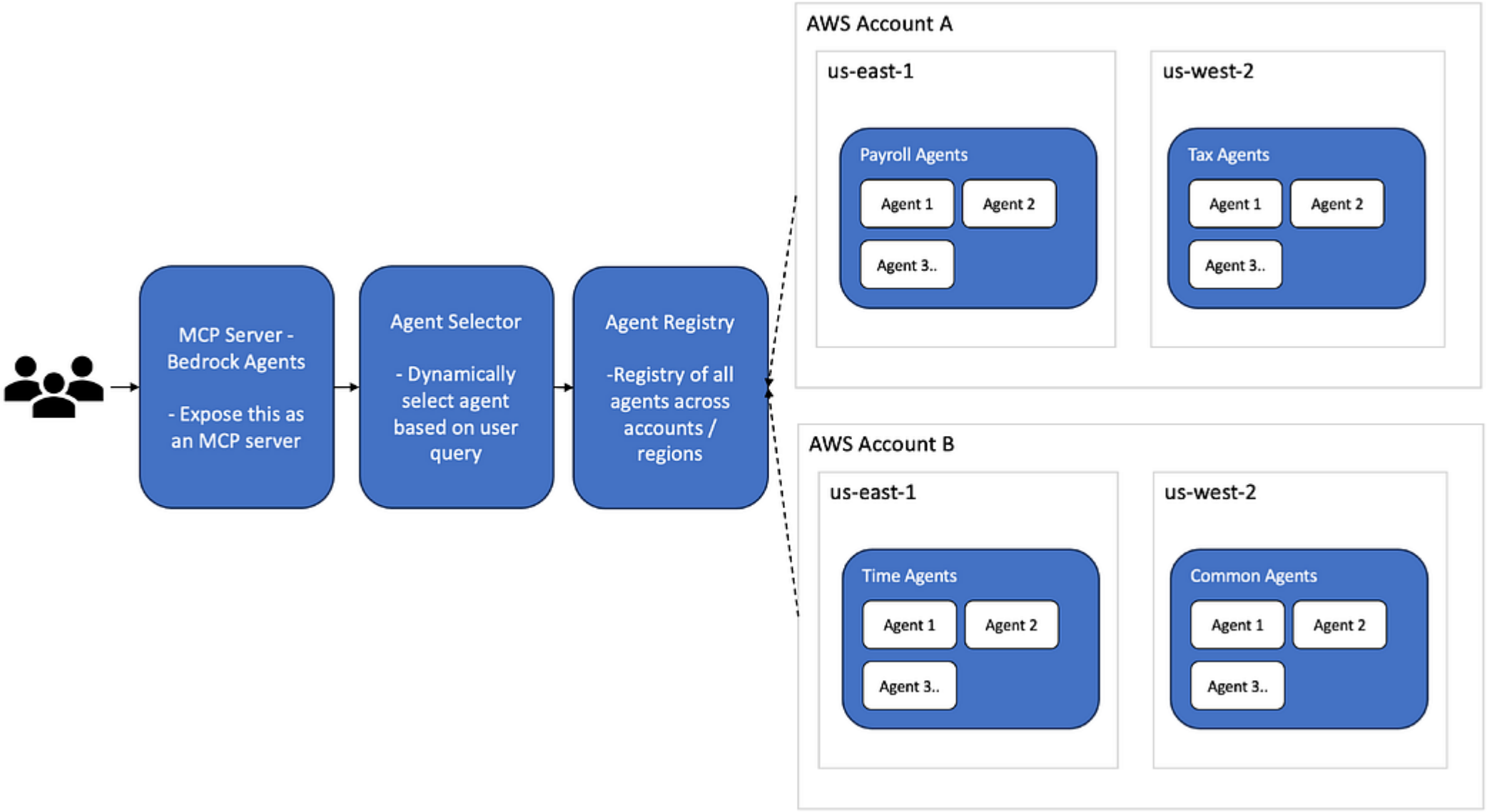
Agent As A Tool



https://meenakshisundaram-t.medium.com/implementing-agent-verse-from-agents-to-agent-registry-to-agent-runtime-to-agent-verse-984335763611

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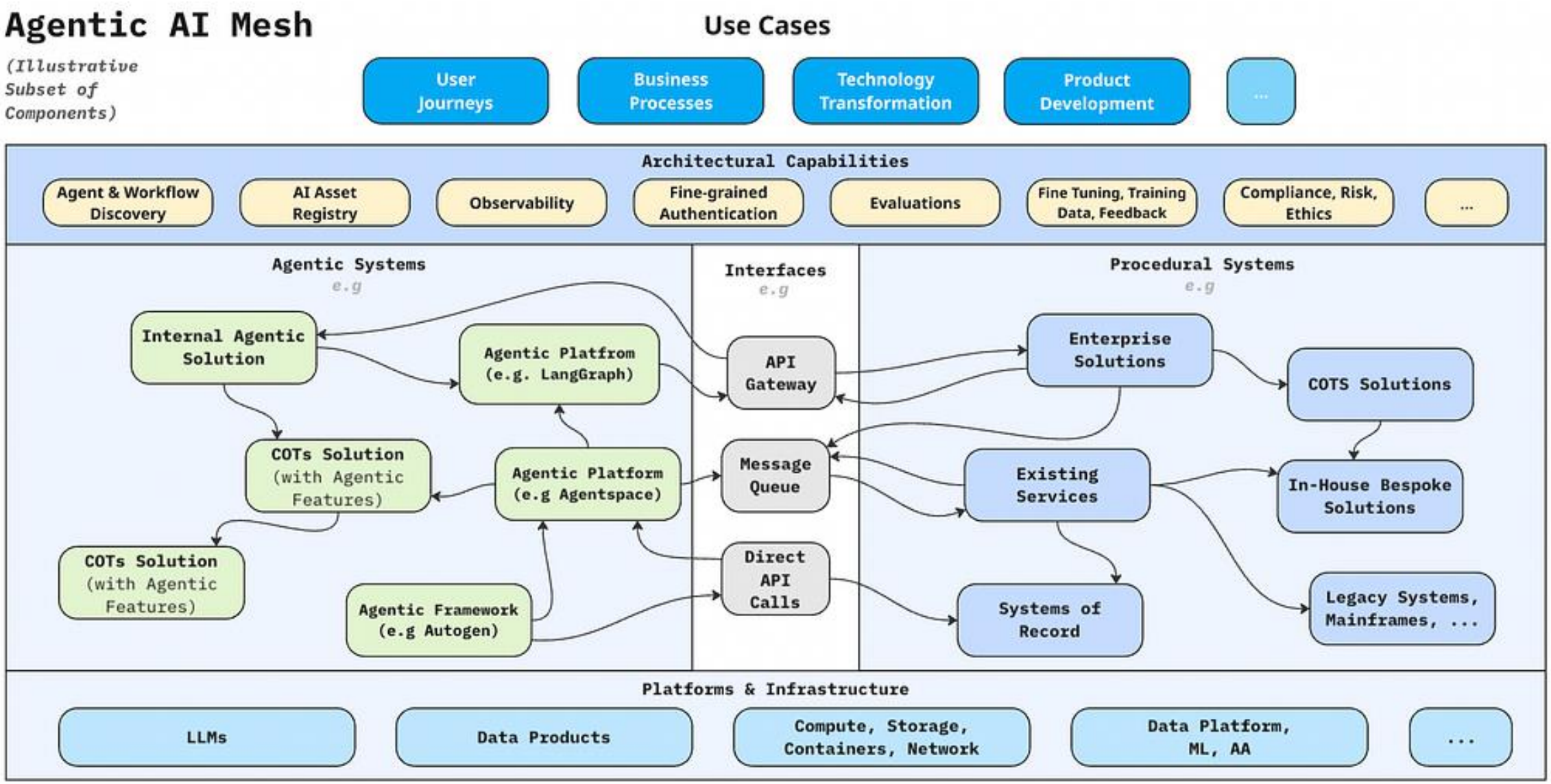
How Communication Messages Flow